

Verified bounds for positive cubic relaxation gaps

Abstract

For multilinear polynomials of degree at most three with nonnegative coefficients, we compare the sum of the individual monomials' envelope widths with the width of the full graph hull. Over all finite nonnegative boxes, the supremum of this ratio lies between $1610000/743033$ and $31/12$. The upper bound uses a fixed mixture of three probability laws that preserves every coordinate marginal and works for all terms at once. Its weights are optimal within that three-law family for a uniform termwise guarantee. An explicit integer-coefficient family gives the lower bound through a rational Bernstein certificate. The accompanying Lean project connects both arguments to the actual graph hulls and includes exact finite witnesses. The exact supremum remains undetermined.

1 Scope and definitions

Let $B = \prod_{i=1}^n [\ell_i, u_i]$ with $0 \leq \ell_i \leq u_i < \infty$, and let

$$f(x) = \sum_{e \in E} a_e m_e(x), \quad m_e(x) = \prod_{i \in e} x_i, \quad a_e \geq 0, \quad |e| \leq 3,$$

where E consists of distinct supports. Constant and affine terms are allowed. Write $\text{vex}_B g$ for the greatest convex underestimator of g on B , and $\text{cav}_B g$ for its least concave overestimator. We compare

$$\begin{aligned} H_B(f; x) &= \text{cav}_B f(x) - \text{vex}_B f(x), \\ T_B(f; x) &= \sum_{e \in E} [\text{cav}_B(a_e m_e)(x) - \text{vex}_B(a_e m_e)(x)]. \end{aligned}$$

Thus T_B concerns the exact hull of each *original* term. Always $0 \leq H_B \leq T_B$. Define $R(3)$ as the supremum of T_B/H_B over this class, arbitrary finite dimension, and points with $H_B > 0$. Fixed coordinates, boundary points, and zero coefficients are allowed.

The comparison of multilinear graph hulls and termwise relaxations is studied by Luedtke et al. [1], who also establish the common concave envelope for positive terms and a factor-two bound for positive bilinear functions. Our definitions use exact individual envelopes on each box; they do not identify them with every recursive McCormick formulation on a general positive box. This paper consolidates the repository's cubic bounds and their formal verification. It does not claim a completed literature-priority assessment.

Theorem 1. *For every polynomial and point above,*

$$T_B(f; x) \leq \frac{31}{12} H_B(f; x).$$

Moreover,

$$\frac{1610000}{743033} = \frac{4830000}{2229099} \leq R(3) \leq \frac{31}{12}.$$

2 Graph hulls and common laws

On the cube, separate affinity gives

$$\text{conv}\{(x, f(x)) : x \in [0, 1]^n\} = \text{conv}\{(v, f(v)) : v \in \{0, 1\}^n\}.$$

Consequently the lower and upper envelopes at x are the minimum and maximum of $\mathbb{E}f(X)$ over binary laws with $\mathbb{E}X_i = x_i$. These extrema are attained. For a nonempty support e , its individual endpoints are

$$\text{cav } m_e(x) = \min_{i \in e} x_i, \quad \text{vex } m_e(x) = \max \left(0, \sum_{i \in e} x_i - |e| + 1 \right).$$

The empty support is constant one. The common-threshold law $X_i = \mathbf{1}_{\{U \leq x_i\}}$, with U uniform on $[0, 1]$, attains all upper endpoints at once. Hence $\text{cav } f = \sum_e a_e \text{cav } m_e$. For any other law P with these marginals, define its deficiency by

$$D_P(e) = \min_{i \in e} x_i - \mathbb{E}_P m_e(X) \geq 0.$$

If one law has $D_P(e) \geq \alpha H(m_e; x)$ for every nonempty support, then $H(f; x) \geq \alpha T(f; x)$. The law must be shared by all terms; choosing a separate law for each term does not give this conclusion.

3 The universal upper bound

Define three global laws with the prescribed marginals:

- O : draw a common uniform U and, independently conditional on U , sample each coordinate with success probability

$$p_O(x_i, U) = \begin{cases} \frac{1}{2} \mathbf{1}_{\{U \leq 2x_i\}}, & x_i \leq 1/2, \\ 1 - \frac{1}{2} \mathbf{1}_{\{U \leq 2(1-x_i)\}}, & x_i > 1/2. \end{cases}$$

This is the folded orientation law used in the formal construction.

- I : draw independent Bernoulli coordinates of means x_i .
- B : call i low when $x_i \leq 1/2$. A low coordinate is $X_i = \mathbf{1}_{\{U \leq x_i\}}$. For each high coordinate, independently conditional on U , set $X_i = 0$ with probability $1/2$ when $U \leq 2(1 - x_i)$, and with probability zero otherwise.

For B , each high coordinate has failure probability $\frac{1}{2} 2(1 - x_i) = 1 - x_i$; all three laws preserve all means. Use the mixture

$$P = \frac{18}{31} O + \frac{6}{31} I + \frac{7}{31} B. \tag{1}$$

3.1 Cubic terms

Order the three means as $u \leq v \leq w$ and put $T = \min(u, 2 - v - w)$. Write D_O, D_I, D_B for the deficiencies. If $v \leq 1/2$, integrating the conditional product for the first two coordinates in O gives $\mathbb{E}_O[X_1 X_2] = u/2$. Thus $D_O \geq u/2$. Also $D_I = u(1 - vw) \geq u/2$. Here $T = u$, so $18D_O + 6D_I + 7D_B \geq 12T$.

If $u \leq 1/2 < v$, put $a = 1 - v$ and $b = 1 - w$, so $a \geq b \geq 0$. Integration of the conditional success probabilities gives

$$D_O = \frac{1}{2} \min(u, a) + \frac{1}{4} \min(u, b), \quad D_B = \frac{1}{2} \min(u, 2a) + \frac{1}{4} \min(u, 2b). \quad (2)$$

We claim $18D_O + 7D_B \geq 12 \min(u, a + b)$. If $a \leq u/2$, the left side is $16a + 8b \geq 12(a + b)$. If $b \geq u/2$, it is at least $18(3u/8) + 7(3u/4) = 12u$. In the remaining cases $b \leq u/2 \leq a$. For $a \leq u$ the left side is $9a + 8b + 7u/2$. When $a + b \leq u$, $3a + 4b \leq 7u/2$; when $a + b \geq u$, $9a + 8b \geq 8u + a \geq 17u/2$. Both give the claim. For $a \geq u$ the left side is $25u/2 + 8b \geq 12u$. The boundaries are included, and $D_I \geq 0$ completes the mixture estimate.

If $u > 1/2$, put $c = 1 - u$, $a = 1 - v$, $b = 1 - w$. Then $1/2 > c \geq a \geq b \geq 0$. Under B , the failure-union probability is $c + a/2 + b/4$, by integration over the three nested failure intervals. Direct integration for O gives the same deficiency:

$$D_O = D_B = a/2 + b/4 \geq 3(a + b)/8 \geq 3T/8.$$

Independence gives the stronger estimate $D_I \geq 7T/16$. Indeed, for $s = a + b \leq 1$ we have $ab \leq s^2/4$ and $D_I \geq u(s - s^2/4)$. If $s \leq u$, then $D_I/T \geq u(1 - u/4) \geq 7/16$. If $s \geq u$, then $D_I/T \geq s - s^2/4 \geq u - u^2/4 \geq 7/16$. These arguments are needed only for $T > 0$; at $T = 0$ nonnegativity suffices. Thus

$$18D_O + 6D_I + 7D_B \geq \left(25\frac{3}{8} + 6\frac{7}{16}\right) T = 12T.$$

3.2 Bilinear terms and boxes

For two means $u \leq v$, $T = \min(u, 1 - v)$ and $D_O = T/2$. If both are low, $D_I \geq T/2$. If just one is low, $D_B = \min(u, 2(1 - v))/2 \geq T/2$. If both are high, $D_B = T/2$ and $D_I = uT \geq T/2$. The mixture therefore captures at least $12T/31$ in every case. Constant and singleton terms have zero gap. Summing the common-law inequalities proves the cube bound.

For a nonnegative box, substitute $x_i = \ell_i + (u_i - \ell_i)y_i$. Each original positive monomial expands into nonnegative cube monomials of degree at most three. The sum of the expanded term widths is at least the original term width: taking a supremum of a sum is bounded by the sum of suprema, and taking an infimum has the reverse inequality. The full hull width is preserved by the affine change of coordinates. This remains true with zero-width coordinates. Apply the cube result to the expansion to prove the upper bound in Theorem 1.

4 Optimality of the fixed mixture

This section concerns uniform termwise guarantees for fixed mixtures of O, I, B , not the exact value of $R(3)$. Let their nonnegative weights be w, r, v with sum one. Suppose their mixture captures at least a fraction α of every bilinear and cubic monomial gap. Three tests give the following normalized deficiencies:

Means	Parameter	$(D_O, D_I, D_B)/T$ or its limit
$(1/2, 1/2)$	fixed	$(1/2, 1/2, 0)$
$(\varepsilon, 1 - \varepsilon^2, 1 - \varepsilon^2)$	$\varepsilon \downarrow 0$	$(3/8, 0, 3/4)$
$(u, 1 - u/2, 1 - u/2)$	$u \downarrow 1/2, u > 1/2$	$(3/8, 7/16, 3/8)$

In the second test take $0 < \varepsilon \leq 1/8$, so $T = 2\varepsilon^2$. The exact independent normalized deficiency is $\varepsilon - \varepsilon^3/2$. In the third test take $1/2 < u \leq 5/8$, which keeps all coordinates high and sorted; $T = u$, and the independent normalized deficiency is $u - u^2/4$. These are tests of the actual laws, with the threshold equality classified as low in the first test. It follows that

$$\alpha \leq \frac{1}{2} - \frac{v}{2}, \quad \alpha \leq \frac{3}{8} + \frac{3v}{8} - \frac{3r}{8}, \quad \alpha \leq \frac{3}{8} + \frac{r}{16}.$$

Average these inequalities with weights $3/31, 4/31, 24/31$. The coefficients of r and v cancel, giving $\alpha \leq 12/31$. Equation (1) attains that guarantee. This does not exclude a better law, coefficient-dependent weights, or an analysis that does not demand the same deficiency fraction separately for every term.

5 An analytic family for the lower bound

Let $m \geq 9$ be a multiple of nine. Use disjoint groups U, V, W of m variables each, with coordinate sums A, B, C . Write $E_k(G)$ for the elementary symmetric polynomial of degree k on a group G . Define

$$f_m = 2E_3(W) + 3BE_2(W) + 2mE_2(V) + \frac{5m}{3}AC + \frac{10m}{9}AB + mE_2(U). \quad (3)$$

Every included term has degree two or three and a positive integer coefficient. Set the group means to $1/4, 1/2, 3/4$, respectively.

5.1 A rational scalar certificate

Put

$$F(a, b, c) = 6c^3 + 27bc^2 + 18b^2 + 30ac + 20ab + 9a^2, \\ L(a, b, c) = 37a + \frac{79}{2}b + 38c - \frac{103}{3}, \quad \delta = \frac{901}{120000}.$$

Lemma 2. For $(a, b, c) \in [0, 1]^3$, $F(a, b, c) \geq L(a, b, c) + \delta$.

Proof. Define

$$p(c) = \frac{-972c^4 + 576c^3 + 1404c^2 - 768c + 101}{96}, \\ q(c) = \frac{-78732c^4 + 212256c^3 - 203796c^2 + 82032c - 11275}{2976}.$$

For $c \leq 3/10$, let $b_* = 13/24 - 3c^2/4$ and $g(c) = -49/6 + 30c - 15c^2 \leq 0$. The identity

$$F - L - p(c) = 9\left(a - 1 + \frac{10}{9}(b - b_*)\right)^2 + \frac{62}{9}(b - b_*)^2 + g(c)(a - 1)$$

proves $F - L \geq p(c)$ when $a \leq 1$. For every real a, b, c ,

$$F - L - q(c) = 9\left(a + \frac{10}{9}b + \frac{30c-37}{18}\right)^2 \\ + \frac{62}{9}\left(b + \frac{9(27c^2 - 79/2) - 10(30c - 37)}{124}\right)^2.$$

It remains to bound p on $[0, 3/10]$ and q on $[3/10, 1]$. For each row below, set $t = (c - \ell)/(u - \ell)$. The indicated polynomial equals $D^{-1} \sum_{i=0}^4 v_i \binom{4}{i} t^i (1 - t)^{4-i}$.

Polynomial	$[\ell, u]$	D	$(v_0, v_1, v_2, v_3, v_4)$
p	$[0, 1/5]$	60000	(63125, 39125, 20975, 9395, 4133)
p	$[1/5, 3/10]$	240000	(16532, 6008, 1802, 3788, 11597)
q	$[3/10, 1/2]$	7440000	(215357, 1285415, 1434125, 1250375, 1008125)
q	$[1/2, 3/4]$	190464	(25808, 18056, 7964, 9230, 15869)
q	$[3/4, 1]$	190464	(15869, 22508, 34520, 45920, 31040)

The Bernstein basis is nonnegative and sums to one. Every v_i/D is at least δ , proving the lemma by exact polynomial identities. $\square$

5.2 From counts to graph hulls

At a binary vertex, put $a = A/m$, $b = B/m$, $c = C/m$. Expansion gives

$$\frac{18}{m^3} f_m = F(a, b, c) - \frac{18c^2 + 27bc + 18b + 9a}{m} + \frac{12c}{m^2} \geq F(a, b, c) - \frac{72}{m}. \quad (4)$$

Every coupling of the prescribed individual marginals has expected normalized counts $(1/4, 1/2, 3/4)$. Taking expectations in Lemma 2 and (4) therefore yields

$$\frac{18}{m^3} \text{vex } f_m \geq \frac{139}{6} + \delta - \frac{72}{m}.$$

Counting the six support classes and using the individual endpoints gives

$$\begin{aligned} \frac{18}{m^3} \text{cav } f_m &= \frac{167}{4} - \frac{153}{4m} + \frac{9}{m^2}, \\ \frac{18}{m^3} T(f_m) &= \frac{161}{4} - \frac{135}{4m} + \frac{6}{m^2}. \end{aligned}$$

Only the $E_3(W)$ terms have nonzero individual lower endpoints. The actual hull gap is positive, and the preceding inequalities give

$$\frac{T(f_m)}{H(f_m)} \geq \frac{161/4 - 135/(4m) + 6/m^2}{223/12 - \delta + 135/(4m) + 9/m^2}. \quad (5)$$

Positivity of the hull gap follows, for example, by comparing the common upper law with independence: every nonlinear support here has means strictly between zero and one and a positive coefficient. As m tends to infinity through multiples of nine, the right side of (5) tends to

$$\frac{161/4}{223/12 - 901/120000} = \frac{1610000}{743033}.$$

Every member belongs to the class defining $R(3)$, so this proves the lower bound in Theorem 1. Dropping the slack gives the simpler bound $483/223$. Already the unrefined finite certificate at $m = 36$ exceeds two. No finite member is claimed to attain either limiting constant, and no limit of the actual ratios is asserted.

6 Existing finite witnesses

The formal distribution also includes exact finite examples. For the two-group family, with both groups of size m , use

$$g_m = AE_2(W) + \frac{5m}{4} E_2(U), \quad x_U = 1/2, \quad x_W = 3/4.$$

For the three-group family, use group means $(1/4, 1/2, 3/4)$ and coefficients $\gamma = (\gamma_0, \dots, \gamma_5)$ on the six terms $(E_3(W), BE_2(W), E_2(V), AC, AB, E_2(U))$.

Family	Variables	Coefficients or parameter	Exact T/H
Two groups	8	$m = 4$	27/16
Two groups	16	$m = 8$	21/11
Two groups	24	$m = 12$	99/50
Two groups	32	$m = 16$	135/67
Three groups	18	$(2, 3, 9, 10, 7, 7)$	20891/10411
Three groups	24	$(2, 3, 13, 12, 8, 7)$	6601/3225
Three groups	192	$(2, 3, 120, 105, 70, 63)$	7443345/3445256

Each exact value is certified by an affine lower bound valid on every integer count vector, a probability law attaining that bound with all individual marginals preserved, and the common upper law. The last four ratios exceed two. These finite certificates are independent of the asymptotic argument above; their full data and kernel proofs are in the accompanying source distribution.

7 Formal verification and boundaries of the result

The standalone project in [formal/](#) contains the dependency closure of the canonical proofs. Its [claim map](#) identifies declarations for the graph-hull interpretation, the three actual laws, scalar inequalities, all-box transfer, fixed-mixture obstruction, finite analytic family, limit and supremum, and finite witnesses. The [verification record](#) gives the pinned Lean and Mathlib versions, commands, module and declaration counts, axiom audit, kernel replay, and source fingerprints. Source export identity and the paper build are checked separately. Standard axioms `propext`, `Classical.choice`, and `Quot.sound` are allowed; unfinished proofs and additional axioms are rejected.

Formal proofs establish the stated mathematical claims about the specified relaxations. They do not establish publication priority, determine the exact value of $R(3)$, or cover every recursive relaxation. General coefficient removal, homogeneous approximation of the supremum, equal-marginal classification, and the general two-group asymptotic family are separate results outside this paper. The finite two-group examples above require none of those extensions.

References

- [1] James Luedtke, Mahdi Namazifar, and Jeff Linderoth. Some results on the strength of relaxations of multilinear functions. *Mathematical Programming*, 136(2):325–351, 2012. doi: 10.1007/s10107-012-0606-z. Author manuscript available at <https://jlinderoth.github.io/papers/Luedtke-Namazifar-Linderoth-12-TR.pdf>.